Design and feasibility evaluation of an LLM-enabled workflow for data capture from paper-based health records

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Abstract. Large language models (LLMs) offer a promising approach to extracting data from paper-based clinical records and providing an independent source for verifying manually entered data. This paper evaluates the feasibility of an LLM-enabled optical character recognition (LLM-OCR) framework and a layout-matching electronic interface designed to support efficient data capture and human-assisted verification. This pilot study evaluated data capture methods using 425 paper-based medical records from 14 healthcare facilities across five districts in Malawi. We developed an electronic data entry interface mirroring the layout of the original records and compared its efficiency with KoboToolbox. Five multimodal LLMs extracted data directly from scanned records, and their outputs were compared with two independently entered human datasets (H1 and H2). Agreement between LLM-OCR and H1 was assessed against the human double-entry baseline (H2 versus H1). The layout-matching interface reduced mean manual entry time from approximately 45 to 13 minutes per record. Human double-entry disagreement was 9% across 410 successfully matched records. Among the five LLMs, GPT-5.6-Terra achieved the highest agreement with H1 (87.4%), followed by Qwen 3VL (77.9%) and GPT-4.1-mini (77.8%), while GPT-4o-mini (51.9%) and NVIDIA Nemotron (43.0%) performed less well. GPT-5.6-Terra achieved 92% agreement with the H2-H1 consensus. Performance varied by field type, e.g. handwritten dates, categorical selections and repeatedly corrected entries presented challenges. Although LLM-OCR achieved lower agreement with human-entered data than conventional double entry, the results demonstrate its potential as an independent source for identifying data entry discrepancies. Combined with a layout-matching electronic interface that facilitates the inspection and correction of extracted values, LLM-OCR provides a promising basis for a human-assisted data verification workflow.

Keywords: LLM-OCR, double-entry agreement, data capture, medical records

1 Introduction

Data capture from paper-based patient or hospital records is an important activity in clinical and epidemiological research. These processes are labour-intensive, expensive and susceptible to human error [7]. Recommended good practice involves double data entry and/or the use of validation and verification processes incorporated into electronic data capture tools [23]. Double data entry may be difficult to implement in studies with limited funding [18]. Reliance on single data entry, even when using electronic systems, increases the risk of transcription errors and information loss. Significant data quality issues have already been documented in settings such as Malawi for both paper-based records and electronic systems [8]. Electronic data capture tools such as KoboToolbox, REDCap, and ODK are popular with researchers as they offer free or low-cost options, but they do not reduce personnel costs for manually entering or verifying data [2, 11, 17].

Automatic data capture through optical character recognition (OCR) can reduce workload and transcription errors when combined with specially designed data collection forms [7, 10, 23]. In the medical domain, OCR has been used to extract specific entities from free text clinical narratives [22], to digitise COVID-19 assessment [30] or laboratory forms [16], to read measurements from medical device screens in ICU settings [20], to extract clinical features from typewritten pathology records [19] or to digitise pharmacy prescriptions [27].

While automatic methods reduce human effort, their exact agreement varies significantly. Some studies found that OCR error rates were not necessarily smaller than those for single data entry [28, 29]. OCR performs better for simpler layouts, structured formats and typewritten documents. To improve their exact agreement on handwritten text or complex layouts, advanced architectures that combine multiple Natural Language Processing and deep learning techniques have been developed [5, 15, 16].

More recently, large language models (LLMs) have been used to support data capture from electronic health records [4, 26, 31]. LLMs seem to offer an attractive alternative for small- to medium-scale studies because they require little or no task-specific model training, are straightforward to deploy through affordable commercial APIs, and may reduce development costs compared to bespoke OCR and information extraction systems. While LLMs do not yet outperform existing state-of-the-art OCR models, they are showing increasingly strong capabilities for recognising handwritten text, particularly English text, and for interpreting complex table structures [3, 9]. Of note is their ability to automatically learn from and use contextual information from documents, thus going beyond the capabilities of traditional OCR tools in analysing layout and detecting the order of data in documents. However, the use of pre-trained general-purpose LLMs for document understanding and structured information extraction remains an open research challenge because models can produce transcription errors, omit information, map fields incorrectly, or even fabricate values [25].

This pilot work focuses on extracting data from paper-based patient records using an LLM-OCR workflow that addresses a rich set of image interpretation challenges: detection of tables, handwritten text such as dates and numbers, circled responses, and

structured extraction of data. This work was undertaken to support data collection for a PhD study investigating early infant HIV diagnosis and linkage to care in Malawi.

1.1 Objectives

The main objective was to develop and evaluate an LLM-OCR workflow for data capture from paper-based patient records that enabled human-assisted verification. A second objective was to assess the efficiency of a layout-matching electronic data entry interface and the potential time and cost savings of the proposed workflow.

2 The Data

In Malawi, Prevention of Mother-to-Child Transmission (PMTCT) "pink cards" are used to monitor children aged 0-24 months born to HIV infected mothers [13]. They constitute the main medical record and a key source of information for the identification, retention and treatment of HIV infections among infants and children [12, 14]. As these records remain largely in paper form, their analysis must capture and convert them into usable electronic data [32].

A pink card has two pages with a well-structured format. The first page contains a top section with child and guardian details, maternal and child enrolment information, and HIV tests. The row with child identifiers and birth cohort information (month and year) is repeated on page two. A table with 24 visit rows captures 15 independent observations per visit, such as the month when the child attended the visit, visit date, height, weight, MUAC, breastfeeding status, ART status of the mother, child HIV test type and results, clinical monitoring, treatment, follow-up dates and outcome. The mother and guardian sections were not captured. In total, a card had 12 variables with handwritten values other than dates, 17 are categories selected by circling, and 11 were handwritten dates. The number of possible independent observations on each card follows the equation:

$$21 + 5\,T + 15\,V \tag{1}$$

where T = number of tests, and V = number of visits. For example, a card with three HIV tests and ten visits would record 186 observations.

3 Methods

Two methods for human data entry were developed. One used a KoboToolbox form, and the other an HTML-based form-aligned interface that closely mimicked the layout of the paper record. Figure 1 shows the data entry processes and agreement evaluations.

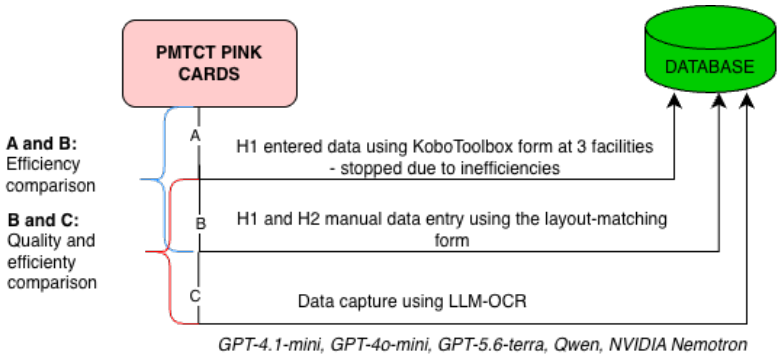


Fig. 1. The data entry processes and agreement evaluations.

The full data collection covered 16 facilities; two facilities were excluded from this pilot due to unavailability of good quality card scans. Initially, H1 entered data manually using the KoboToolbox at the first five facilities in one district. The form was long on multiple pages, required extensive scrolling and frequent interaction with dropdown menus, making navigation cumbersome and time-consuming. To improve data entry time, a new form that mimicked the layout of the paper-based form was developed. A subset of the data entered with KoboToolbox was entered again using with the layout-matching form by H1 to compare data entry time. As this proved effective, H1 continued entering data with the new form for the remaining 11 facilities. H2 independently entered data using the layout-matching form for all facilities.

3.1 Data entry using the KoboToolbox form

The KoboToolbox, given in the supplementary material, had two sections for each patient. A registration section captured identifying information of the respondent and generated a unique respondent identifier. The second section contained visit fields and was dynamic, allowing subsequent visits to be added. The respondent section and the visits section were linked using the respondent identifier and a visit number thereby preserving the longitudinal relationship between records. When an HIV-test sample was recorded as collected during a visit, the corresponding test was entered into the HIV-testing register table, while the test result was recorded both against the corresponding visit row and separately in the HIV-testing table. Although this approach provided a logical way of linking HIV-test data to visits using KoboToolbox, it also involved some interpretation and corrections being applied. When missing or inconsistent HIV test data was found, H1 inferred missing the values and corrected dates. As a result, some values in the electronic entered data reflected decisions made by H1 during data entry rather than information directly documented on the original form. This affected agreement for a few of the HIV-tests fields.

3.2 Data entry with the layout-matching form

We developed a new form that matched the layout of the paper form (Fig. 2). Field groupings and field order were preserved. Unlike the KoboToolbox form, it displayed the form sections all on one page, allowing users to visually compare them with the original record. Incorrectly entered visit rows could be more easily identified and corrected. Electronic fields allowed values to be entered as they appeared on the cards, without requiring conversion to a standardised format.

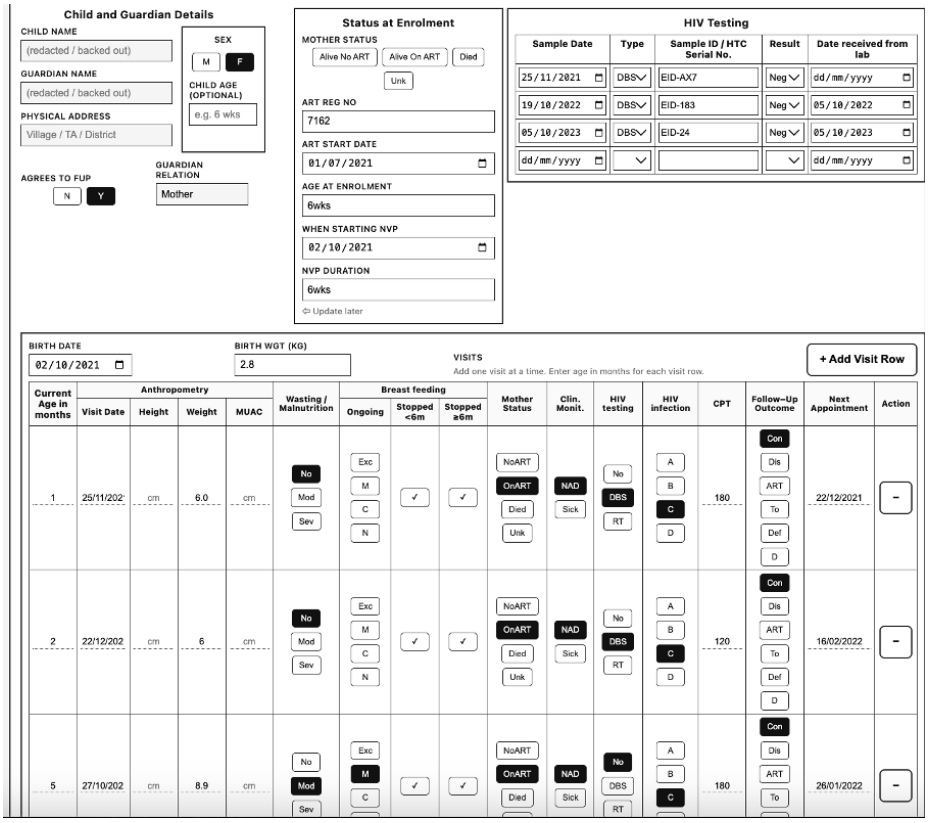


Fig. 2. Layout-matching electronic interface form used to enter data from a pink card. Image shows a completed electronic form visually matching the physical form layout and format.

3.3 Data capture using LLM-OCR

LLM-OCR was used to extract data from scanned cards using five LLMs, including two open-weight, locally deployable models. Cards were submitted as one file with both pages. All records were processed once. Prompts were identical for all models.

Automated batch processing extracted data without human intervention, saved the outputs as JSON and CSV files, and consolidated them at facility level. Records were matched using facility registration numbers, and field-level exact agreement.

GPT models were accessed through the OpenAI API, which did not submit data for model training, used AES-256 encryption at rest and TLS 1.2 or higher in transit. Qwen and NVIDIA Nemotron were deployed through AWS Bedrock, with zero operator access and data retention, KMS encryption at rest and TLS 1.2 in transit.

All scanned images excluded sensitive fields for the guardian and mother's details. The health centre card registration number was read by the OCR and used for card matching within each facility. This was a facility-level identifier not a nationally unique patient identifier.

GPT-5.6-Terra used Top-p=0.98, temperature=1 and a low reasoning effort. GPT-4.1-mini and GPT-4o-mini both used temperature=1 and Top-p=1. Qwen3 VL 235B A22B and NVIDIA Nemotron Nano 12B v2 VL BF16 were chosen as locally deployable options, both used temperature of zero.

3.4 Agreement measures

Cards were matched across datasets using the facility-level registration number handwritten in a designated box at the top of each card. Visit-level agreement was assessed after aligning LLM-extracted visit rows with the corresponding human-entered visits, using the age month of the child at the visit. Unmatched visits were recorded separately from field-level disagreements. Exact field-level agreement was calculated only for eligible aligned visits, using populated fields in the reference dataset.

To evaluate the quality of the LLM-OCR extracted data, we calculated exact field agreement. H1 was used for reference because H1 had more expertise in HIV data. To establish a human baseline, H2 was compared against H1. Agreement between LLM-OCR and H1 and a three-way agreement against the H2 - H1 consensus were calculated. Agreement was measured as the proportion of values that were identical in the comparison datasets. For each pairwise comparison, one dataset was the reference and the other was the candidate. A reference-blank addition was an eligible, aligned field for which the candidate recorded a value and the reference was blank. The false positive rate (FPR) was the number of reference-blank additions divided by the total number of blank reference fields compared. The unsupported rate (UR) was the same count but divided by the total number of populated candidate fields compared. Values were normalised before exact comparison, e.g. dates and numbers put into a consistent format.

4 Ethical approval

The study was approved by the Malawi University of Business and Applied Sciences Research Ethics Committee (MUBAS REC) (Approval No. RI-2025-0054). Permission to access paper-based HIV patient records was granted by health authorities and participating facilities. The study involved a retrospective review of routinely collected clinical records and no direct patient contact. The ethics committee waived the requirement

for individual informed consent because only existing records were used. Before data extraction, all direct patient identifiers were obscured, and only de-identified data were processed by the OCR workflow. Data were stored securely and were accessible only to authorised members of the research team.

5 Results

The analysis dataset contained 425 cards from 14 facilities in five districts of Malawi which were independently entered by H1 and H2. This was a subset of the entire data collected because we excluded two facilities as these were among the first entered using the KoboToolbox; one of these facilities had no scans and the other had poor-quality images. Data, code and all supplementary materials are at https://github.com/avtaylor/llm-ocr-paper.

5.1 Time saving

On average, each record contained 9–11 visits, and 2 HIV tests, corresponding to 166-196 potential observations per card. The time required to enter a card using the KoboToolbox form was approximately 45 minutes. This meant that the target of a minimum 20 cards per facility could not be completed in one day. We found that the new form simplified error correction and reduced the effort required to review entered data. Data entry time was reduced to an average of 13 minutes per card, saving an average of 30 minutes per card compared to KoboToolbox.

5.2 Agreement between H2 and H1

Of the 425 cards, 410 were successfully matched between the two human-entered datasets. The remaining fifteen cards were not matched automatically because of discrepancies in the recorded registration numbers arising from transcription errors or illegible identifiers on the original cards. All subsequent agreement analyses were restricted to the 410 matched cards. Human disagreement across matched cards was 9%. Cells left blank by H2 where H1 had entered a value are counted as disagreement. Agreement for circled categorical indicators was over 95%, except for the follow-up field which had more categorical options available for selection. Agreement for date fields ranged from 78% to 90%. Many date discrepancies involved differences in a single digit. The FPR was 3.4%, while the UR was 0.6%.

5.3 LLM-OCR exact agreement against human data entry

We calculated agreement for LLM-OCR on the 410 cards matched between H1 and H2. Model-specific analyses were further restricted to cards successfully matched by the corresponding LLM-OCR outputs, resulting in different numbers of matched cards and visits across models. The exact agreement of the LLM-OCR was between 43.0% and 87.4%. GPT-5.6-Terra achieved the highest agreement rate with H1 among the

models evaluated (**Table 1**). NVIDIA Nemotron showed the lowest overall agreement. Qwen performed comparably to GPT-4.1-mini. Overall, GPT-5.6-Terra achieved between 88–95% agreement for circled selections, followed by Qwen. Nemotron frequently interpreted all printed options as selected and read many empty visit rows. Qwen extracted some additional sparsely populated visit rows, including those containing only partial information, such as a visit date, where the entry had subsequently been abandoned or corrected in another row. The human data entrants recognised and did not enter these rows.

Table 1. Agreement between LLM-OCR and H1. The number of matched visits, agreement and disagreement rates are calculated for successfully matched cards. The denominator was the number of H1-populated matched cells. FPR and UR used their own defined denominators. Exact Agg. = exact agreement; Diss. = disagreement; FPR = false positive rate; UR = unsupported rate.

Model vs H1	Matched Cards	Matched Visits	Exact Agg. %	Diss. %	FPR %	UR %
GPT-4.1-mini	396	2,897	77.8	22.2	24.7	4.1
GPT-4o-mini	395	1,208	51.9	48.1	33.4	6.8
GPT-5.6-Terra	397	2,935	87.4	12.6	8.2	1.4
Qwen 3VL	387	2,918	77.9	22.1	25.4	4.3
NVIDIA Nemotron	397	2,827	43.0	57	30.1	5.4

NVIDIA Nemotron showed low agreement for circled categorical fields (24.8%), compared with 68.0–71.9% for the other field types. This suggests that interpreting circled selections was an important source of disagreement for Nemotron. GPT-4o-mini matched a similar number of cards to GPT-5.6-Terra (395 versus 397), but substantially fewer visit rows (1,208 versus 2,935). A close inspection indicated that GPT-4o-mini frequently omitted visit rows or failed to align extracted rows with the corresponding human-entered visits. This contributed to its lower field-level agreement.

Three-way exact agreement. The three-way exact agreement was calculated as the percentage of comparable cells where all three H1, H2 and LLM-OCR record the same normalised value. The denominator was the number of comparable cells populated by both H1 and H2 (note the difference to the denominator used in **Table 1**). A cell left blank by LLM-OCR counted as a disagreement when both H1 and H2 supplied a value. A cell where either H1 and H2 left blank was outside of the denominator.

Table 2. Pairwise and three-way agreement between H1, H2 and LLM-OCR (%). The denominator comprises aligned cells in which both H1 and H2 entered a value. The H1–H2 column reports agreement between the two human-entered datasets on these cells.

Model	H1 vs H2	Model vs H1	Model vs H2	Three-way agreement	Model vs consensus
GPT-4.1-mini	95.3	79.2	79.2	77.4	81.3
GPT-4o-mini	95.1	51.8	51.8	50.8	53.5
GPT-5.6-Terra	95.3	89.5	89.3	87.6	91.9
Qwen 3VL	95.3	79.2	79.0	77.4	81.2
NVIDIA Nemotron	95.2	44.4	44.5	43.1	45.3

In **Table 2** and **Figure 3**, for visit fields, all three compared values referred to the same visit on the card. The model agreement versus human consensus was calculated used the same numerator as the three-way agreement, but the denominator was restricted to cells where both H1 and H2 agreed. H1-H2 agreement was approximately 91% when all eligible H1-populated fields were considered, including omissions by H2; when restricted to aligned fields populated by both, agreement increased to approximately 95%. GPT-5.6-Terra agreed with each H1 and H2 about 89% of the time, roughly 6 percentage points below the H1-H2 benchmark. GPT-5.6-Terra had the best performance leading by about 10 percentage points or more above other models. The model came closest to human consensus and was able to reproduce roughly 92 out of every 100 aligned cells where the two human data entrants agreed.

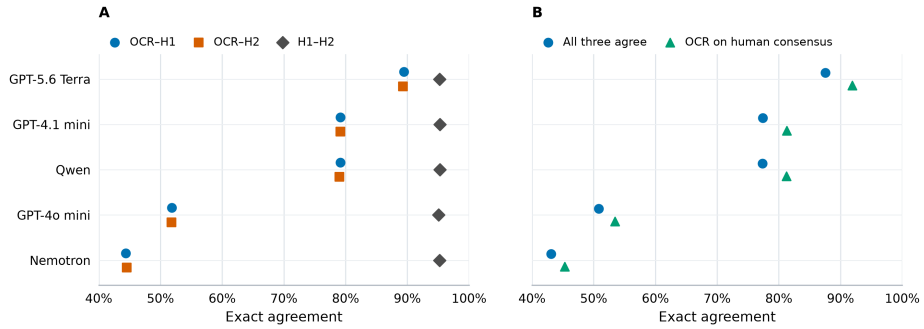


Fig. 3. Pairwise (Panel A) and three-way agreement (Panel B) on 410 eligible H1-H2 cards. Pairwise and three-way agreements used aligned cells populated by both H1 and H2, while consensus agreement used cells where H1 and H2 agreed.

Looking closer at agreement for types of observations (Table 3). Dates were all handwritten and had the lowest agreement in the human entered data. This trend was also observed for LLM-OCR which extracted dates with lower exact agreement. Circled categorical values were extracted with a high exact agreement by GPT-5.6-Terra, while Nemotron performed poorest for this field type. Weight and height measurements generally showed higher agreement than CPT dosages.

Table 3. Exact agreement per field type (%). Human pairwise agreement is directional and uses populated values in the reference dataset. LLM-OCR agreement is against human consensus.

Field Type	Fields	H2 vs H1	H1 vs H2	GPT-5.6-Terra	Qwen 3VL	GPT-4.1-mini	GPT-4o-mini	NVIDIA Nemotron
Categorical circled	17	93.8	97.3	95.4	79.1	79.7	58.2	24.8
Handwritten excl. dates	12	87.1	92.7	88.8	86.1	83.3	51.2	71.9
Dates	11	83.9	86.9	79.7	75.9	77.5	40.8	68.0

Types of errors. An error of substitution occurred when both the reference and the candidate contained values that differed after normalisation. An error of omission occurred

when the reference contained a value and the candidate field was blank. In the human agreement errors were almost equally split between errors of substitution and omission. In contrast, substitutions dominated OCR disagreements with the reference and accounted for 79-94% of disagreements on populated H1 cells across the five models. Qwen had the fewest omissions against the human reference, while Terra had the lowest substitutions and addition rates.

Date fields were a significant source of error for both humans and automatic data extraction. The pattern of H2 and H1 discrepancies was similar to that observed for Terra against H1: about 64% of substitution errors for dates involved one-digit difference. Typical LLM-OCR recognition problems included reading and recording two-digit and four-digit years differently, and confusing closely written digits such as 1/7, 3/8, or 5/6. Use of abbreviations, e.g. 22/Sep/23, and different separators were the source of difference only for a small proportion of dates errors (less than 6% for Terra). Some differences were caused by day/month reversal patterns. An audit against the scanned cards was needed to establish the exact source of disagreement.

Most models read circled fields well even though circles were drawn by hand and were not perfect, often crossing over text. Combined OCR exact agreement for circled values was around 95% for Terra. Two of the fields, breastfeeding and HIV infection status, produced more errors in this group.

During preliminary testing, we observed substantially better extraction agreement when scanned cards were rotated to landscape orientation. We therefore standardised image orientation before processing. This effect was not evaluated in a separate controlled experiment.

5.4 Cost considerations

We spent less than US$1 to extract data from all the cards using the OpenAI models; this included initial tests. The measured mean total card processing time was 68 seconds. This excluded time to load the card in the scanner and was inflated by rendering the image on the screen for verification and by the internet / network speed. Internet speeds in Malawi can be very low and fluctuate widely. A median response time from GPT models for one card was between 15 and 30 seconds. GPT-4o-mini used expensive tile-based accounting due to its method for resizing, dividing and converting images into tokens. This may also have affected its overall performance versus GPT-4.1-mini. Qwen was less computationally intensive than NVIDIA, and achieved a better agreement overall.

5.5 LLM-OCR as a double-entry audit tool

The LLM-OCR process provided an independent source of extracted values that could be compared with manually entered data to identify discrepancies. The layout-matching form facilitated the identification and correction of discrepancies between automatically extracted and human-entered values, compared with KoboToolbox. Categorical values could be entered and corrected quickly by using clickable buttons rather than

requiring manual entry or selections using dropdown menus as was the case for KoboToolbox forms.

LLM-OCR encountered difficulties with fields containing multiple handwritten corrections. In a human-assisted workflow, where records are processed individually, such discrepancies could be identified at the start and resolved by a reviewer before proceeding with data verification.

6 Discussion

We evaluated two complementary components of a hybrid data capture workflow: a layout-matching electronic interface for manual data entry and an LLM-OCR process. The interface substantially reduced manual entry time, while LLM-OCR demonstrated potential to support discrepancy identification and subsequent human verification.

Our electronic data entry form followed proximity-compatibility and control-display compatibility principles to minimise navigation and to reduce the eye movement of the reviewer when locating and entering information from the paper card. Circled values are designed as buttons that become highlighted when selected and/or activated/deactivated for choices that were mutually exclusive. The layout also reduced the review time for dates and numeric fields by balancing between validated and free text entry. Other studies have also shown the importance of consistency in layout and careful positioning of items on the screen to reduce unnecessary navigation [15, 24]. Improvements to these aspects have been shown in other studies to increase data completeness substantially from 30% to 70% [6].

By using the layout-matching form, we showed that data entry time and time needed for verification could be substantially reduced. The initial data entry target at each facility was 20 cards, hence the achieved reduction in entry-time from 45 minutes with KoboToolbox to 13 minutes, the number of cards collected at facilities was increased and data entry at each facility could be completed in one day.

Real-time data entry can be supported in the way we showed. LLM-OCR can be used to pre-populate values and to facilitate double-entry checks. Electronic data entry is most effective when it makes manual review and correction fast and less error-prone [21]. Error-correction is important when moving data between multiple electronic formats. In our experiments, the best LLM-OCR model, GPT-5.6-Terra, produced an exact cell/field agreement with human consensus of over 91% and, for many individual fields of interest, exact agreement was over 95%. While these agreement rates are not directly comparable to those reported for structured form processing, our findings are promising and closely resonate regarding the inherent challenges of extracting handwritten text and checkbox values [21]. Template-driven workflows rely on pre-defined document layouts and error rates are calculated on values extracted from these known areas; therefore, errors are typically lower. However, the manual completion of OCR-ready templates requires more effort and time than more human-ready forms [1]. LLM-OCR has the advantage that it does not require specially designed forms or pre-defined reading boxes. It is likely that the LLM models identified the tabular structure during

the prompt construction and testing. Specifying the output format as structured JSON may have also helped guide the models in this task.

Most substitution errors made by the LLM-OCR were one- or two-character mistakes and could be corrected efficiently using the layout-matching form. Mistakes of omission were fewer. We noted that LLM-OCR struggled to read handwritten circled values. These also pose difficulties for traditional OCR tools [29]. Dates and numeric values were also problematic as they were handwritten using many formats and styles.

Our study has limitations. This was a pilot study and used a small sample. We measured agreement between LLM-OCR outputs and independently entered human data rathers than agreement against an adjudicated reference dataset. Consequently, not all disagreements could be attributed to errors in the LLM-OCR output, as human transcription or interpretation errors may have also contributed. Conversely, agreement between human entrants did not guarantee that the recorded value was correct. Further evaluation against verified values from the original scanned records is needed to establish extraction accuracy and the effectiveness of LLM-OCR-assisted verification.

Secondly, we relied entirely on the LLMs to identify and interpret the form layout without providing bounding boxes or regions of interest. Further improvements in the LLM-OCR may also be achieved by incorporating document layout and semantic rules. We could provide examples to enable the GPT to auto-correct or recognise certain errors, or to read only the final corrected values for crossed-out fields.

Thirdly, we applied minimal image pre-processing to simulate realistic scenarios. The only pre-processing step applied, which substantially improved performance, was rotating images to landscape, to match the orientation of the physical cards. We found that exact agreement dropped for cards that were bent, old and worn out.

The batch process used JPEG images which were smaller than the larger PNG format which could be created when processing cards individually. The use of JPEG images was required to meet batch processing input-size limits. The lossy compression may have obscured small handwriting, decimal points, circled selections and date digits compared with lossless PNG images.

Data privacy is an important consideration when using LLMs. There is an important trade-off between the benefits of LLMs and the need to safeguard the confidentiality of patient records. In settings such as Malawi, where records are mostly paper based, LLM-OCR could help achieve record keeping at an affordable cost while allowing for measures to preserve the privacy and security of patient data. Segments of protected personal information such as names, physical addresses, and unique patient numbers tied to health data can be blacked out and kept outside of the data capture pipeline. Running the LLM-OCR entirely within a healthcare institution's secure infrastructure would further reduce concerns related to external data transmission and third-party processing. However, local deployment requires good computing infrastructure, model maintenance, and security controls. Exact agreement for open-source or open-weights models should be evaluated against state-of-the-art commercial systems.

7 Conclusion

We provide practical insights into automating data capture and verification from health records using LLMs. Our findings demonstrate that a layout-matching electronic interface can substantially reduce manual data entry time for structured paper-based clinical records. Multimodal LLMs also showed potential to provide an independent source of extracted values for comparison with human-entered data, although agreement varied considerably across models and field types. These results support the feasibility of a hybrid workflow combining automated extraction with human verification. Further evaluation is required to establish whether this approach reduces verification effort and improves final data accuracy compared with conventional double data entry.

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